



Machine Learning

Course (Category) Code	Course Name	Teaching Scheme (Hrs/week)					Credits Assigned			
		L	T	P	O	E	L	T	P	Total
(PCC)	Machine Learning	2	0	2	4	8	2	0	1	3
		Examination Scheme								
		Component		ISE		MSE		ESE		Total
		Theory		13		14		40		67
CE307		Laboratory		26		--		7		33

Pre-requisite Course Codes, if any.	
Course Objective: To learn methodology and tools to apply machine learning algorithms to real data and evaluate their performance.	
Course Outcomes (CO): <i>At the End of the course, students will be able to</i>	
CE307.1	Define the fundamental principles of Machine learning.
CE307.2	Apply techniques, mathematical concepts, and algorithms used in machine learning.
CE307.3	Interpret the limitations of various machine learning algorithms and the way to evaluate the performance of machine learning algorithms
CE307.4	Design an application through software implementation of different concepts and algorithms covered in the course.

CO-PO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CE307.1	2	2	--	--	--	--	--	--	2	2	2	2
CE307.2	2	2	2	--	2	--	--	--	2	2	2	2

CE307.3	2	2	2	--	2	--	--	--	2	2	2	2
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CE307.4	2	2	2	2	2	--	--	--	2	2	2	2
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CO-PEO/PSO Correlation Matrix (3-Strong, 2-Moderate, 1-Weak Correlation)

	PEO1	PEO2	PEO3	PEO4	PSO1	PSO2	PSO3
CE307.1	2	2	1	1	1	--	2
CE307.2	2	2	1	1	2	--	2
CE307.3	2	2	1	1	2	--	2
CE307.4	2	2	1	1	2	1	2

BLOOM'S Levels Targeted (Pl. Tick appropriate)

Remember	Understand	Apply	Analyze	Evaluate	Create✓
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Theory Component

Module No.	Unit No.	Topics	Ref.	Hrs.
1	Title	Introduction to Machine Learning	T1	04
	1.1	Machine Learning, Key terminology, Key tasks of machine learning, How to choose the right algorithm, Types of Machine Learning, Issues in Machine Learning, Application of Machine Learning, Steps in developing a Machine Learning Application.		
2	Title	Learning with Trees	T1, T2, T3	06
	2.2	Learning with Trees: Decision Trees, Constructing Decision Trees using Gini Index (Regression), Classification and Regression Trees (CART)		
	2.3	Performance Metrics: Confusion Matrix, [Kappa Statistics], Sensitivity, Specificity, Precision, Recall, F-measure, ROC		
		curve		

3	Title	Learning with Support Vector Machine	T1, T2	05
	3.1	Support Vector Machine Constrained Optimization, Optimal decision boundary, Margins and support vectors, SVM as a constrained optimization problem, Quadratic Programming, SVM for linear and nonlinear classification, Basics of Kernel trick.		
	3.2	Support Vector Regression, Multiclass Classification		
4	Title	Ensemble Learning		06
	4.1	Understanding Ensembles, K-fold cross-validation, Boosting, Stumping, XGBoost	T2	
	4.2	Bagging, Subbagging, Random Forest, Comparison with Boosting, Different ways to combine classifiers	T2	
5	Title	Learning with Clustering and Dimensionality Reduction		7
	5.1	Introduction to clustering with an overview of distance metrics and major clustering approaches.	T1, T2	
	5.1	Dimensionality Reduction Techniques, Principal Component Analysis, Linear Discriminant Analysis, Singular Valued Decomposition.	T1, T2	
6	Self Study	Graph Based Clustering, Reinforcement Learning	T2	5*
Total				28

Laboratory Component, if any. (Minimum 10 Laboratory experiments are expected)

All experiments should be performed through PYTHON Language

Sr. No	Title of the Experiment
1	Implement Linear Regression of Prediction/ Classification
2	Implementation on Decision Tree learning
3	Implement SVM for linear data
4	Implement SVM for non-linear data using kernel function
5	Implementation on Ensemble learning: Bagging using Random Forest

6	Implementation on Ensemble learning: Boosting using XGBoost / AdaBoost
7	Implement K-means/ K-Modes Clustering
8	Implement Expectation Maximization(EM) algorithm to Find Natural Patterns in Data
9	Implement PCA/SVD for Dimensionality Reduction
10	Implement LDA for Dimensionality Reduction

Text Books

S.N.	Title	Edition	Authors	Publisher	Year
1	Machine Learning In Action	1st	Peter Harrington	DreamTech Press	2001
2	Introduction to Machine Learning	3rd	Ethem Alpaydin	MIT , PRESS	2012
3	Machine Learning	1st Edition	Tom Mitchell	Mc-Grawhill	1997

Reference Books

S.N.	Title	Edition	Authors	Publisher	Year
1	Pattern Recognition and Machine Learning,	1st Edition	C. M. Bishop	Springer	2013
2	Elements of Statistical Learning	2 nd Edition	Trevor Hastie Robert Tibshirani Jerome Friedman	Springer	2001
3	Machine Learning An Algorithmic Perspective		Stephen Marsland	CRC Press	

Useful Digital Links

1	Data sets for Machine Learning algorithms: https://www.kaggle.com/datasets
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2	Machine Learning repository- https://archive.ics.uci.edu/ml/index.php
3	Machine Learning from Coursera
4	https://towardsdatascience.com/machine-learning/home
5	https://onlinecourses.nptel.ac.in/noc21_cs85/preview